

CELL AND MOLECULAR BIOLOGY LABORATORY

From Gene Mutation to Disease

Investigating How DNA Sequence Changes Affect Protein Products and Human Phenotypes

Background

Many human diseases and biological traits are associated with changes in DNA sequence. A mutation may involve one nucleotide or many nucleotides. It may change an amino acid, introduce a premature stop codon, shift the reading frame, remove amino acids, add amino acids, or affect gene function in another way.

In this laboratory, you will investigate one human disease or medically relevant phenotype and one associated gene. You will study the normal function of the gene, obtain its normal coding sequence(CDS), translate it to obtain the predicted normal protein sequence, manually reproduce a documented mutation, translate the mutant sequence, and compare the normal and mutant proteins.

Your main goal is to explain the following relationship: DNA mutation → change in coding sequence→change in protein → change in molecular or cellular function → disease or phenotype.

Learning Objectives

1. Investigate the molecular basis of a human genetic disease or phenotype.
2. Retrieve an appropriate human reference CDS and protein sequence.
3. Explain the normal function of a selected gene and protein.
4. Identify a documented genetic variant associated with a disease or phenotype.
5. Manually reproduce a mutation in a nucleotide sequence.
6. Translate normal and mutant coding sequences.
7. Compare normal and mutant protein sequences.
8. Identify mutation types such as missense, nonsense, frameshift, in-frame insertion/deletion, and repeat expansion.
9. Predict how a mutation may affect protein function.
10. Relate molecular changes to cellular and organismal phenotypes.
11. Distinguish computational prediction from experimental evidence.
12. Document the laboratory workflow using Galaxy and GitHub.

Important Scientific Distinction

Translation of a DNA sequence predicts the protein sequence encoded by that CDS. It does not demonstrate that the protein is actually expressed in a human cell. Protein expression, abundance, localization, folding, and activity require additional experimental evidence. Use the term “predicted protein sequence” when discussing the output of sequence translation.

Part I. Select a Human Disease and Gene

Each student must investigate a different human disease/phenotype–gene combination. Your selected disease and gene must be approved by the instructor. Examples include HIV susceptibility–CCR5, cystic fibrosis–CFTR, sickle cell disease–HBB, Huntington disease–HTT, achondroplasia–FGFR3, phenylketonuria–PAH, Wilson disease–ATP7B, Li-Fraumeni syndrome–TP53, hereditary breast/ovarian cancer–BRCA1 or BRCA2, and familial Parkinson disease–LRRK2.

Part II. Research the Disease

Before changing any sequence, understand the biology of your disease and gene.

A. Disease

13. What is the disease or phenotype?

- Achondroplasia, a genetic skeletal disorder characterized mainly by disproportionate short stature and shortened limbs.

14. What are its major clinical characteristics?

- Major features include disproportionate short stature, rhizomelic shortening of the limbs, macrocephaly, frontal bossing, midface retrusion, brachydactyly, trident-shaped hands, genu varum (bowed legs), exaggerated lumbar lordosis, and thoracolumbar kyphosis. Other possible complications include spinal stenosis, obstructive sleep apnea, and foramen magnum/craniocervical junction problems.

15. Which cells, tissues, or organs are mainly affected?

- The developing skeletal system, particularly chondrocytes in the growth plates, is mainly affected. Abnormal regulation of chondrocyte proliferation and differentiation leads to impaired endochondral bone growth, especially in the long bones. FGFR3 is also expressed in other tissues, but the characteristic phenotype primarily involves skeletal development.

16. What is its genetic basis?

- Achondroplasia is caused by activating (gain-of-function) pathogenic variants in the FGFR3 gene. The most common variant is NM_000142.5:c.1138G>A, which changes glycine to arginine at amino acid 380 (p.Gly380Arg / G380R). This causes excessive FGFR3 signaling and reduces chondrocyte proliferation.

17. What is its inheritance pattern, if applicable?
- Autosomal dominant. About 80% of affected individuals have a de novo FGFR3 variant, meaning it occurs for the first time in the affected individual rather than being inherited from an affected parent.
15. Which cells, tissues, or organs are mainly affected?
- The skeletal system, particularly the growth plates, is mainly affected. Abnormal FGFR3 signaling affects chondrocytes, the cartilage-forming cells involved in endochondral bone growth.
16. What is its genetic basis?
- Achondroplasia is mainly caused by pathogenic variants in the FGFR3 gene. The common c.1138G>A variant causes the p.Gly380Arg amino-acid substitution and results in abnormal activation of FGFR3 signaling.
17. What is its inheritance pattern, if applicable?
- Achondroplasia has an autosomal dominant inheritance pattern. One pathogenic FGFR3 allele is sufficient to cause the condition. Approximately 80% of affected individuals have a de novo FGFR3 variant rather than inheriting it from an affected parent.

B. Gene and Normal Protein

18. What is the official gene symbol?
- FGFR3 — fibroblast growth factor receptor 3.
19. On which human chromosome is the gene located?
- Chromosome 4, short arm (4p16.3).
20. What does the gene normally encode?
- It encodes fibroblast growth factor receptor 3 (FGFR3), a receptor tyrosine kinase located in the cell membrane. The receptor has an extracellular ligand-binding region, a transmembrane segment, and an intracellular tyrosine kinase domain.
21. What is the normal biological function of the protein?
- FGFR3 normally regulates cell proliferation, differentiation, and survival, with an important role in chondrocyte regulation and normal skeletal development. It responds to fibroblast growth factors and activates downstream signaling pathways.
22. Where in the cell is the protein normally found?
- FGFR3 is a cell-surface/transmembrane receptor, with its ligand-binding portion outside the cell and its tyrosine kinase domain inside the cytoplasm.
23. In what biological pathway or cellular process does it participate?

- It participates in fibroblast growth factor (FGF) receptor signaling, including downstream pathways such as RAS–MAPK/ERK, which regulate cell proliferation and differentiation. In bone, this signaling controls chondrocyte activity and endochondral bone growth. The achondroplasia variant causes excessive receptor signaling.

C. Documented Mutation

Variant Information	Details
Gene	FGFR3
Reference transcript	NM_00142.5
Reference protein	NP_000133.1
Exact variant notation	NM_000142.5(FGFR3):c.1138G>A(p.Gly380Arg)
Nucleotide change	c.1138G>A
Codon change	GGC → AGC
Predicted protein change	p.Gly380Arg(G380R)
Nucleotides affected	1 nucleotide: position 1138
ClinVar Variation ID	16327
ClinVar accession	RCV000017724.96
Clinical interpretation	Pathogenic
Associated condition	Achondroplasia

ClinVar identifies c.1138G>A as a missense variant and reports the protein change as p.Gly380Arg. The current ClinVar record shows a Pathogenic germline classification with multiple submitters and no conflicts.

Part III. Obtain the Normal Reference Sequence

The normal reference coding sequence (CDS) for the FGFR3 gene was obtained from the NCBI RefSeq database. The reviewed RefSeq transcript NM_000142.5 represents FGFR3 transcript variant 1 and encodes the reference FGFR3 protein NP_000133.1. The CDS was selected for the manual translation experiment because it represents the

protein-coding portion of the transcript.

Reference Information	Details
Gene	FGFR3
Reference transcript	NM_000142.5
Reference protein	NP_000133.1
Sequence type used	Coding sequence (CDS)
Database	NCBI RefSeq
FASTA filename	FGFR3_WT_CDS.fasta

The normal FGFR3 CDS was saved in FASTA format as FGFR3_WT_CDS.fasta and will be used as the wild-type (WT) sequence for the subsequent translation and mutation experiments.

Part IV. Import the Normal Sequence into Galaxy

Upload or import your WT CDS into Galaxy. Create a new Galaxy history named: Surname_Disease_Gene_Mutation_Lab. Do not place unrelated analyses in this history.

Part V. Establish the Wild-Type Control

The normal sequence is your wild-type (WT) control. Translate the normal CDS into its predicted amino- acid sequence using an appropriate Galaxy translation tool. Save the protein FASTA with a clear filename such as CCR5_WT_protein.fasta.

Record the CDS length, predicted protein length, start codon, stop codon, reading frame, first 10aminoacids, and last 10 amino acids. Compare your predicted protein with the accepted reference protein. If the sequences do not agree, investigate the reason before continuing.

WT Sequence Information	Result
Gene	FGFR3
Reference transcript/CDS	NM_000142. 5

Reference protein	NP_000133.1
CDS length	2421 bp
Predicted protein length	806 amino acids + stop codon
Start codon	ATG
Stop codon	TGA
Reading frame	Frame 1
First 10 amino acids	MGAPACALAL
Last 10 amino acids	FAHDLLPPAP
Protein FASTA filename	FGFR3_WT_protein.fasta

The WT FGFR3 CDS is 2421 bp and translates into a predicted 806-amino-acid protein plus a stop codon. Translation begins with ATG in reading frame +1, producing the normal FGFR3 protein sequence used as the reference for comparing the mutation..

Part VI. Formulate a Mutation Hypothesis

Before editing the sequence, predict what the documented mutation should do. Record:

- Mutation and exact nucleotide change
- Number of nucleotide(s) affected
- Predicted mutation type
- Predicted effect on the reading frame
- Predicted effect on protein length
- Predicted effect on protein function

Write your prediction before generating the mutant protein.

Mutation Prediction	Expected Result
Mutation	FGFR3 c.1138G>A
Exact nucleotide change	G to A at nucleotide 1138
Number of nucleotides affected	1
Predicted mutation type	Missense

Predicted codon change	GGC to AGC
Predicted amino-acid change	Gly to Arg at position 380
Predicted effect on reading frame	No change
Predicted effect on protein length	No change
Predicted effect on protein function	Expected to alter FGFR3 protein function signaling because one amino acid is changed

Hypothesis:

The FGFR3 c.1138G>A mutation is predicted to be a single-nucleotide missense substitution that changes the codon from GGC to AGC, resulting in a Gly380Arg amino-acid substitution. Because only one nucleotide is substituted, the reading frame and protein length are expected to remain unchanged, but the amino-acid substitution is expected to alter FGFR3 protein function.

Part VII. Create the Mutant Sequence

Make a copy of your WT sequence. Never alter your only WT copy. Give the mutant sequence a clear name, for example CCR5_Delta32_CDS.fasta or BRCA1_c68_69delAG_CDS.fasta.

Using the documented variant information, manually alter the copied nucleotide sequence. Depending on your mutation, this may involve substituting, deleting, inserting, or duplicating nucleotide(s), or modifying a repeat.

You must reproduce the documented mutation. Do not randomly alter the sequence.

Record the original nucleotide position(s), original sequence, mutant sequence, number of bases inserted/deleted/substituted, and mutation type.

Required Information	Result
Original nucleotide position	c.1138
Original sequence	G
Mutant sequence	A
Bases inserted	0
Bases deleted	0
Bases substituted	1 (G to A)

Mutation type	Single-nucleotide substitution/missense mutation
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Part VIII. Translate the Mutant Sequence

Translate the mutant CDS using the same procedure used for the WT sequence. Save the mutant proteinFASTA. Record the mutant CDS length, mutant protein length, reading frame, location of the first amino- acid difference, presence or absence of a premature stop codon, and approximate number of amino acids affected.

Required Information	Result
Mutant CDS length	2421 bp
Mutant protein length	806 amino acids
Reading frame	Frame 1
First amino-acid difference	Gly380 to Arg380
Premature stop codon	Absent
Approximate number of amino acids affected	1 amino acid
Mutant protein FASTA filename	FGFR3_WT_CDS_Mutant.fasta (modified)

Part IX. Compare WT and Mutant Proteins

Compare the WT protein and mutant protein using an appropriate sequence alignment method when needed.

24. At what amino-acid position do the sequences first differ?

- Amino acid 380

25. Is only amino acid affected?

- Yes

26. Are multiple downstream amino acids changed?

- No

27. Was an amino acid deleted or inserted?

- No

28. Was a premature stop codon produced?

- No

29. Did the reading frame change?

- No, it remains in frame 1

30. Did the protein length change?

- No, it remains 806 amino acids

31. Is the mutation missense, nonsense, frameshift, in-frame deletion/insertion, repeat expansion, or another type?

- mutation changes one amino acid (Glycine to Arginine) without changing the reading frame or protein length.

Part X. Explain the Molecular Consequence

The FGFR3 mutation associated with achondroplasia is a single-nucleotide substitution called c.1138G>A. In the computational analysis, the nucleotide G was changed to A, without any insertion or deletion of nucleotides. Therefore, this mutation is classified as a substitution and not as a frameshift or insertion/deletion mutation.

This nucleotide change affects the codon at amino acid position 380, changing Glycine (Gly) into Arginine (Arg), which is known as p.Gly380Arg or G380R. The translated protein sequence showed that the first difference occurred at amino acid 380. Only this amino acid was changed, with no additional amino acid changes, insertions, deletions, or premature stop codons. The reading frame also remained unchanged, and the protein stayed at 806 amino acids long. These results indicate that the mutation is a missense mutation.

The Gly380Arg substitution occurs in the transmembrane region of the FGFR3 receptor. Glycine is a small amino acid, while arginine is larger and positively charged. This change can affect how the FGFR3 receptor is regulated. Studies have shown that the G380R mutation can cause the receptor to become active even without the normal growth-factor signal, resulting in excessive FGFR3 signaling.

The increased FGFR3 signaling affects chondrocytes in the growth plate. In particular, the mutation has been associated with increased MAPK/ERK signaling, which can interfere with normal chondrocyte proliferation and differentiation. Since chondrocytes are important for the growth of cartilage in the growth plate, this abnormal signaling can reduce the normal process of bone growth.

As a result, endochondral ossification is disrupted, which limits the normal longitudinal growth of bones. This eventually contributes to the characteristic skeletal features of achondroplasia, including disproportionate short stature and shortened long bones.

Part XI. Second Experiment: Create Your Own Mutation

Artificial Mutation Information	Result
Gene	FGFR3
Reference transcript	NM_000142.5
Mutation type	Single-nucleotide substitution
Nucleotide position	c.7
Nucleotide change	G to A
Codon change	GCC to ACC
Predicted amino acid change	Alanine to Threonine
Predicted mutation type	Missense
Predicted frameshift	No
Predicted premature stop	No
Predicted protein length	806 amino acids

Part XII. Compare the Documented and Artificial Mutations Compare the WT sequence, documented mutation, and student-created mutation. Include CDS length, protein length, mutation type, whether the reading frame changed, whether a premature stop codon appeared, amino acids affected, and expected functional consequence. Explain why the two mutations produced similar or different outcomes.

Feature	WT	Documented Mutation	Student-Created Mutation
CDS length	2421 bp	2421 bp	2421 bp
Protein length	806 aa	806 aa	806 aa
Mutation	None	c.1138>A	c.7G>A
Codon change	None	GGC to GAC	GCC to ACC
Mutation type	None/wild type	Missense	Missense

Reading frame changed?	No	No	No
Premature stop codon?	No	No	No
Amino acids affected	None	Gly380 to Arg	Ala3 to Thr
Expected functional consequence	Normal FGFR3 Function	Increased FGFR3 signaling associated with achondroplasia	Predicted change in FGFR3 protein due to Ala3 to Thr substitution

Part XIII. Interpretation Questions

32. Why does the exact location of a mutation matter?

- The location determines which codon or amino acid is affected and whether the mutation changes the protein. A mutation in an important functional region may have a greater effect than one in a less critical region.

33. Why can deleting three nucleotides produce a different result from deleting one or two nucleotides?

- Deleting three nucleotides removes one complete codon, so the reading frame stays intact. Deleting one or two nucleotides shifts the reading frame, changing the codons and amino acids downstream.

34. Does every mutation change the amino-acid sequence? Explain.

- No. A mutation can be synonymous, meaning the changed codon still codes for the same amino acid. Mutations can also occur in noncoding regions and therefore may not directly change the protein sequence.

35. Does every amino-acid substitution destroy protein function? Explain.

- No. Some substitutions have little or no effect on protein function, while others can alter protein structure or activity. The effect depends on the amino acid and its location in the protein.

36. Why can a frameshift affect many amino acids even if only one nucleotide was deleted?

- Deleting one nucleotide changes how the remaining nucleotides are grouped into codons. Therefore, every codon after the deletion can be read differently, potentially changing many amino acids and producing a premature stop codon.

37. Why might a premature stop codon produce a nonfunctional protein?

- A premature stop codon causes translation to end too early, producing a shortened protein that may lack important regions required for its structure or function.

38. Could a mutation affect protein function without greatly changing protein length?

- Yes. A single-nucleotide substitution can change one amino acid while leaving the protein length unchanged. Your artificial FGFR3 c.7G>A mutation is an example: the protein remains 806 amino acids long but changes Ala3 to Thr3.

39. Could a mutation cause disease without changing the protein sequence? Give a possible molecular mechanism.

- Yes. A mutation could affect gene regulation, such as changing a promoter, enhancer, or splice-regulatory sequence. This could alter how much or when a normal protein is produced without changing its amino-acid sequence.

40. What evidence from your analysis supports the proposed molecular mechanism of your disease?

- Our analysis showed that the documented FGFR3 c.1138G>A mutation changes the coding sequence from GGC to GAC, producing a Gly380Arg amino-acid substitution without changing the reading frame or protein length. This supports a mechanism involving an altered FGFR3 protein rather than a frameshift or truncated protein. The computational analysis alone does not establish exactly how that substitution changes receptor activity.

41. Which conclusions are supported directly by your computational results, and which require evidence from published experimental studies?

- The computational results directly support the DNA sequence change, codon change, amino-acid change, reading-frame status, and predicted protein length. Claims about the mutation's actual effect on FGFR3 activity and how that altered activity causes achondroplasia require experimental and published biological evidence beyond the sequence analysis.

Github Link: <https://github.com/jerricktayos05/Tayos-FGFR3-Mutation-Lab.git>

Part XIV. GitHub Documentation

Your GitHub repository will serve as your electronic laboratory notebook. Do not upload large genomic datasets. Use the following simple organization:

```
Disease-Gene-Mutation-Lab/  
  README.md  
  01_reference/  
    WT_CDS.fasta  
    WT_protein.fasta  
  02_documented_mutation/  
    mutant_CDS.fasta  
    mutant_protein.fasta  
  03_artificial_mutation/  
    artificial_mutant_CDS.fasta  
    artificial_mutant_protein.fasta  
  04_results/  
    WT_vs_mutant_alignment.txt  
    results_summary.md  
  05_report/  
    final_report.md
```

Your README.md must identify your name, disease/phenotype, gene, reference transcript accession, reference protein accession, documented variant, ClinVar accession when available, Galaxy historyname, and date of analysis.

Part XV. GitHub Commit History

Your repository must show the progression of your work. Do not upload everything in one final commit. Suggested progression:

- Commit 1: Initialize disease-gene project
- Commit 2: Add WT reference CDS and protein
- Commit 3: Document disease and normal gene function
- Commit 4: Add documented mutation
- Commit 5: Add mutant protein and comparison
- Commit 6: Add artificial mutation experiment
- Commit 7: Add interpretation and final report

Your GitHub history forms part of the evidence that you performed the activity.

Part XVI. Final Report

Your final_report.md should contain:

42. Title
43. Disease Background
44. Gene and Normal Protein Function
45. Documented Mutation
46. Hypothesis
47. Methods
48. Results
49. WT versus Mutant Protein Comparison
50. Artificial Mutation Experiment
51. Molecular Interpretation: gene → mutation → protein → cellular effect → phenotype 52.
- Limitations
53. Conclusion
54. References

Required Submission

55. GitHub repository URL
56. Galaxy history evidence as instructed by the instructor
57. README.md
58. WT CDS
59. WT protein
60. Documented mutant CDS
61. Documented mutant protein
62. Artificial mutant CDS
63. Artificial mutant protein
64. WT-mutant comparison/alignment
65. final_report.md
66. Scientific references

Important Rules

- Do not modify the original WT sequence. Always work on a copy.
- Do not invent a disease mutation. Your first mutation must be supported by a reliable genetic or medical source.
- Do not randomly delete nucleotides and call the result a known mutation. Reproduce the documented sequence change.
- Do not confuse predicted protein sequence with experimentally demonstrated protein expression.
- Do not upload large unnecessary files to GitHub.
- Document every manual sequence edit clearly.
- Most importantly: do not report only what changed. Explain why the change matters biologically.

